

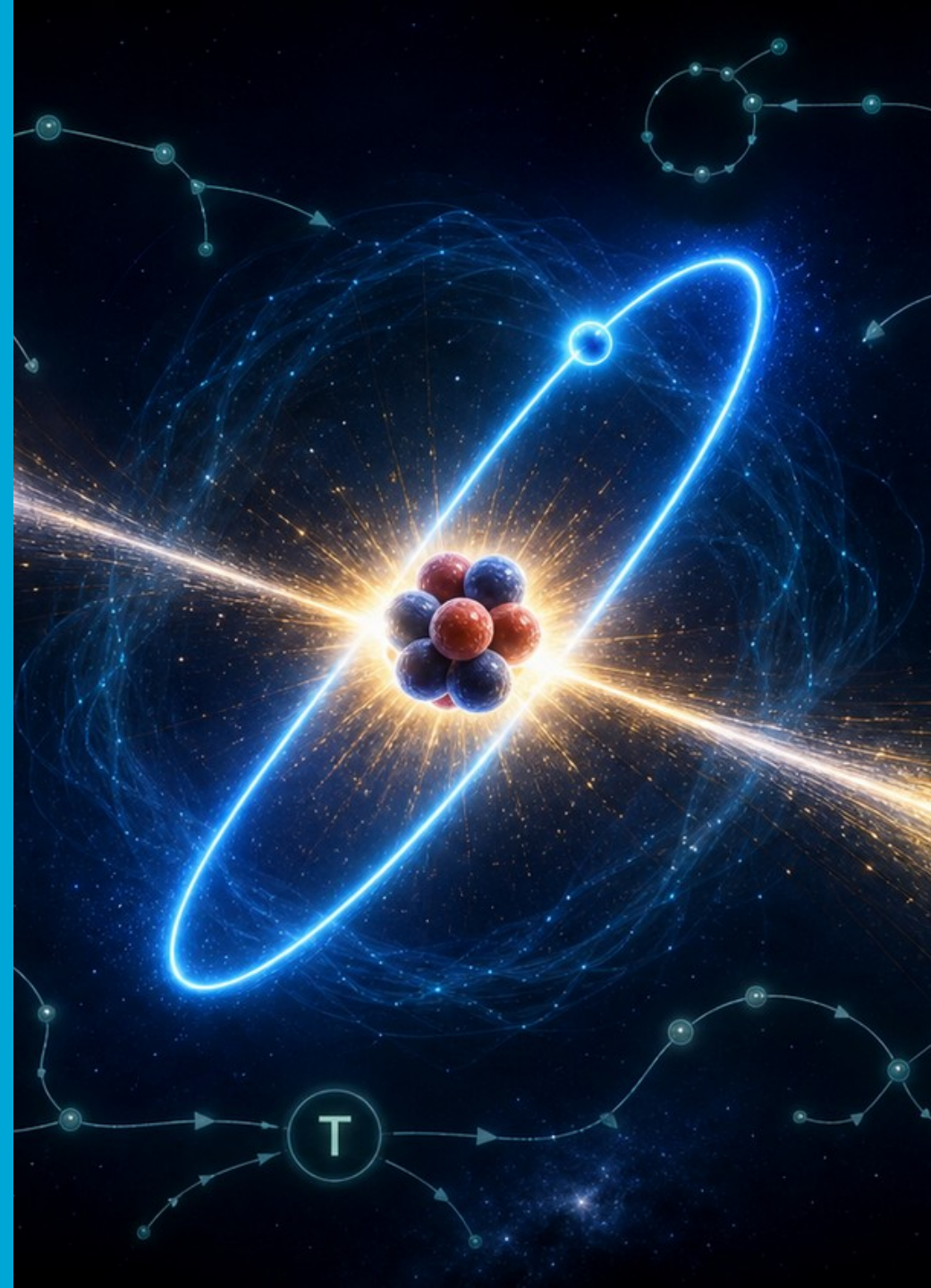
Cold fusion

Can muons break the ice?

Simulating
Muon Catalysed Fusion (μ CF)
using Kinetic Monte Carlo (KMC)



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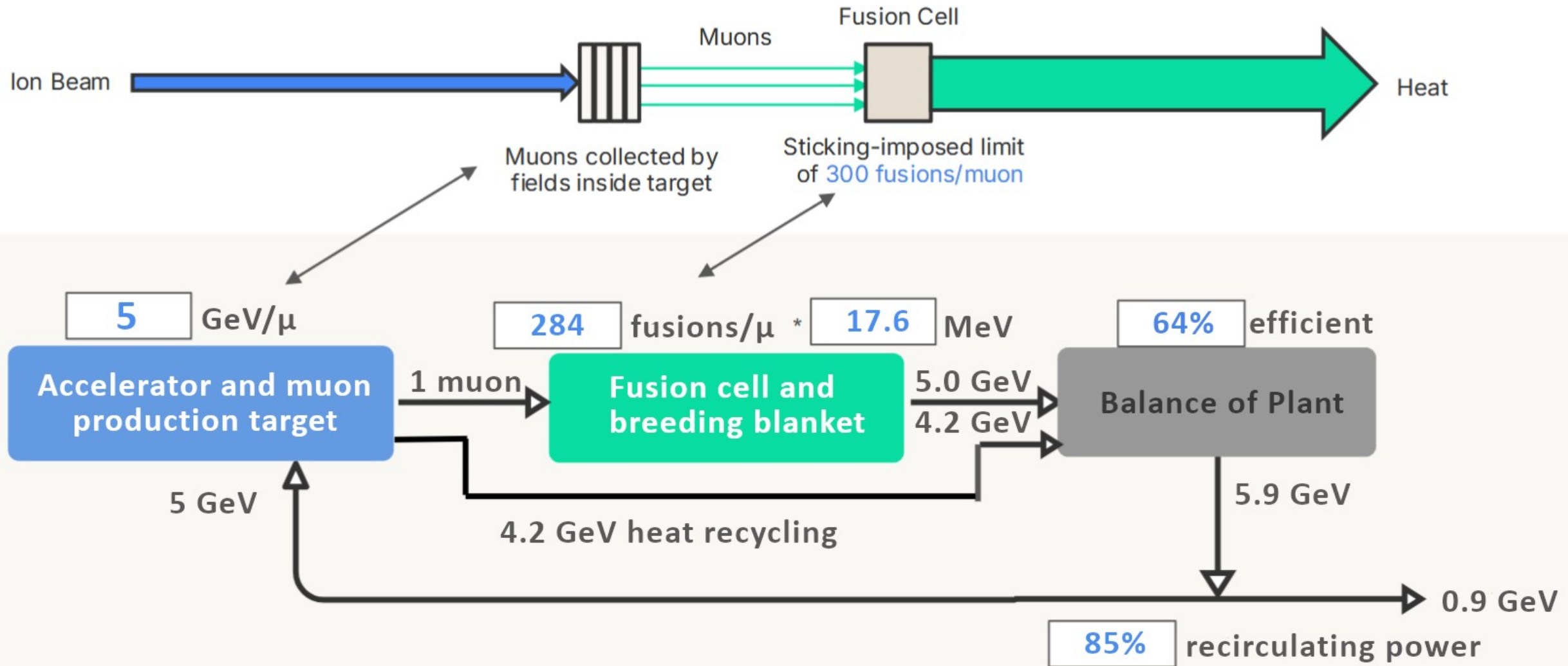
A decorative graphic on the left side of the slide. It features a large white number '0' centered on a blue background. The blue background is partially covered by green, wavy, organic shapes that resemble liquid or smoke. The entire graphic is set against a dark blue background that covers the right half of the slide.

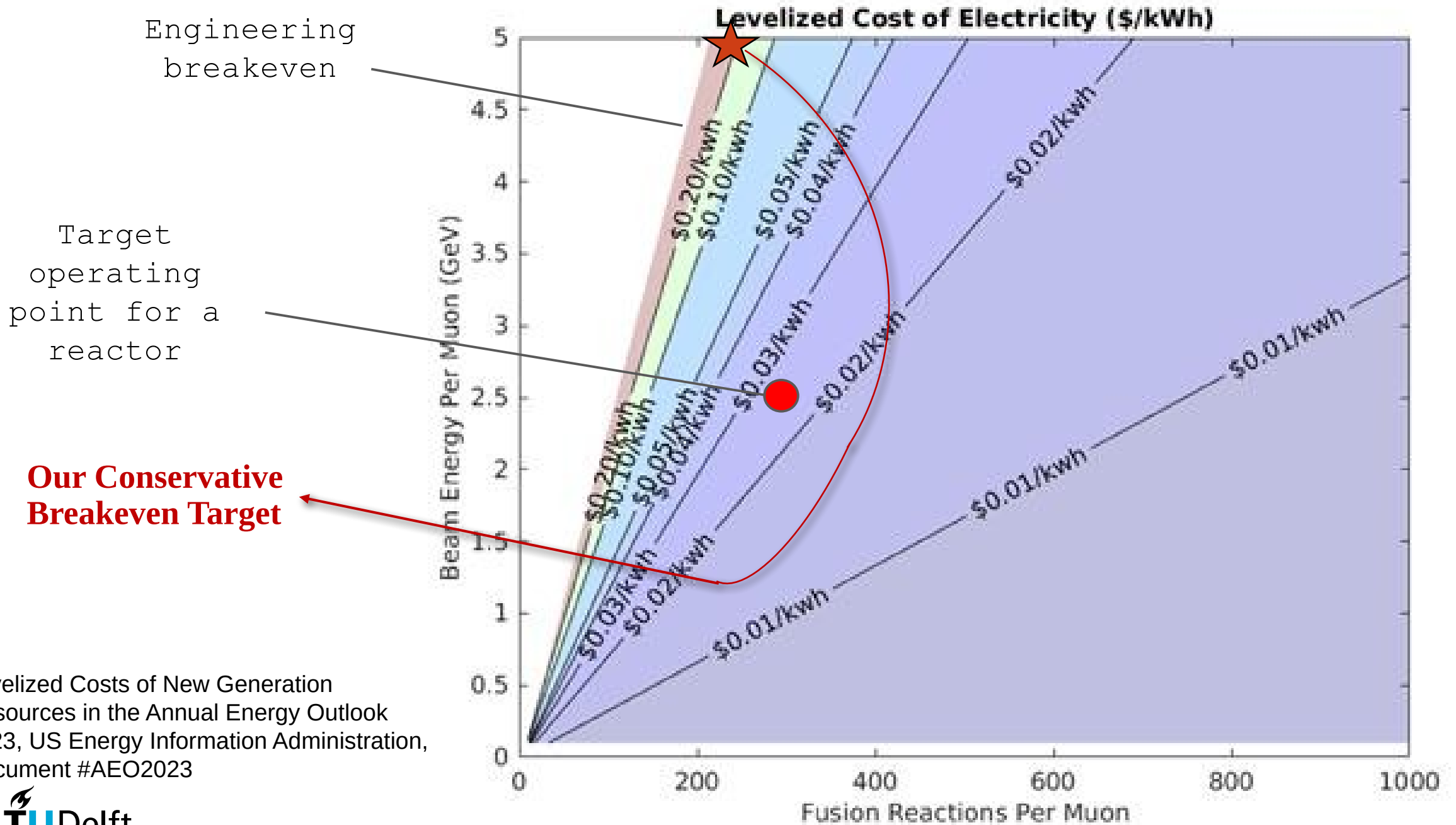
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Motivation

Turning serendipity into success!

Near term μ CF-Reactor Feasibility





A decorative graphic on the left side of the slide. It features a large white number '1' centered on a blue background. The blue background is partially covered by green shapes, including a large curved shape on the left and a smaller one on the right.

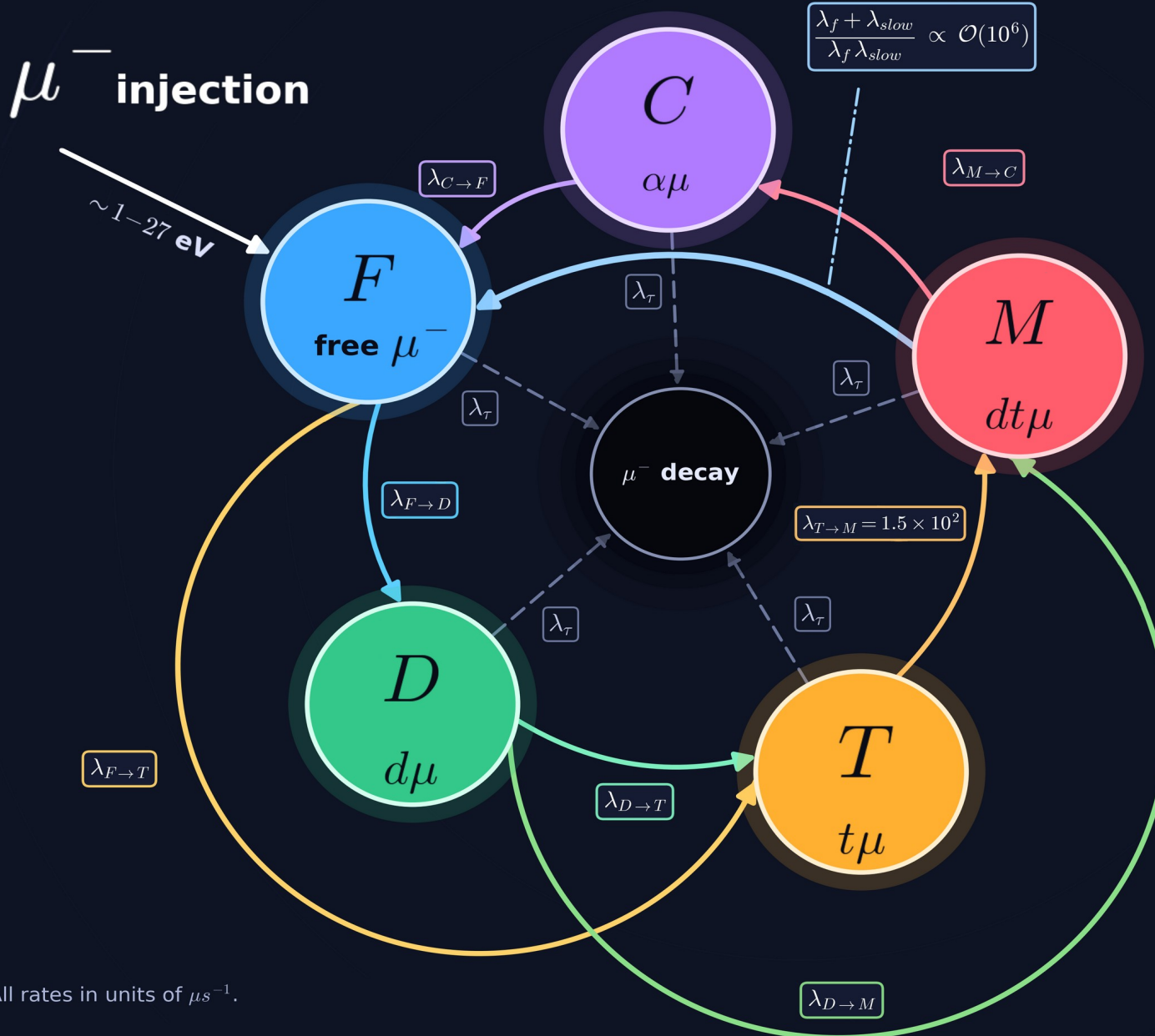
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Introduction

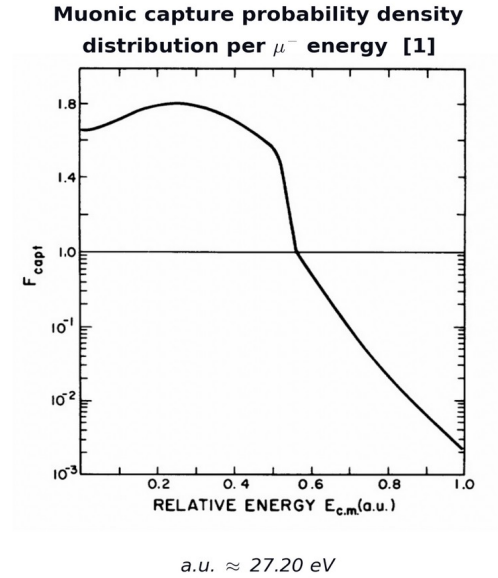
Kinetic Monte Carlo simulation

μ CF Dynamics in Deuterium-Tritium

Muon-Catalyzed Fusion Cycle in D-T: KMC State Diagram



$\lambda_{F \rightarrow D}, \lambda_{F \rightarrow T}$ sampled from:



States

-  F : free muon slowdown
-  D : muonic atom $d\mu$
-  T : muonic atom $t\mu$
-  M : muonic molecule $dt\mu$
-  C : alpha-stuck muon $\alpha\mu$

$\lambda_\tau = 0.455$ **all decays $\rightarrow$ hub**



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Validation

Can KMC reproduce the reduced model?

Reduced Rate Model

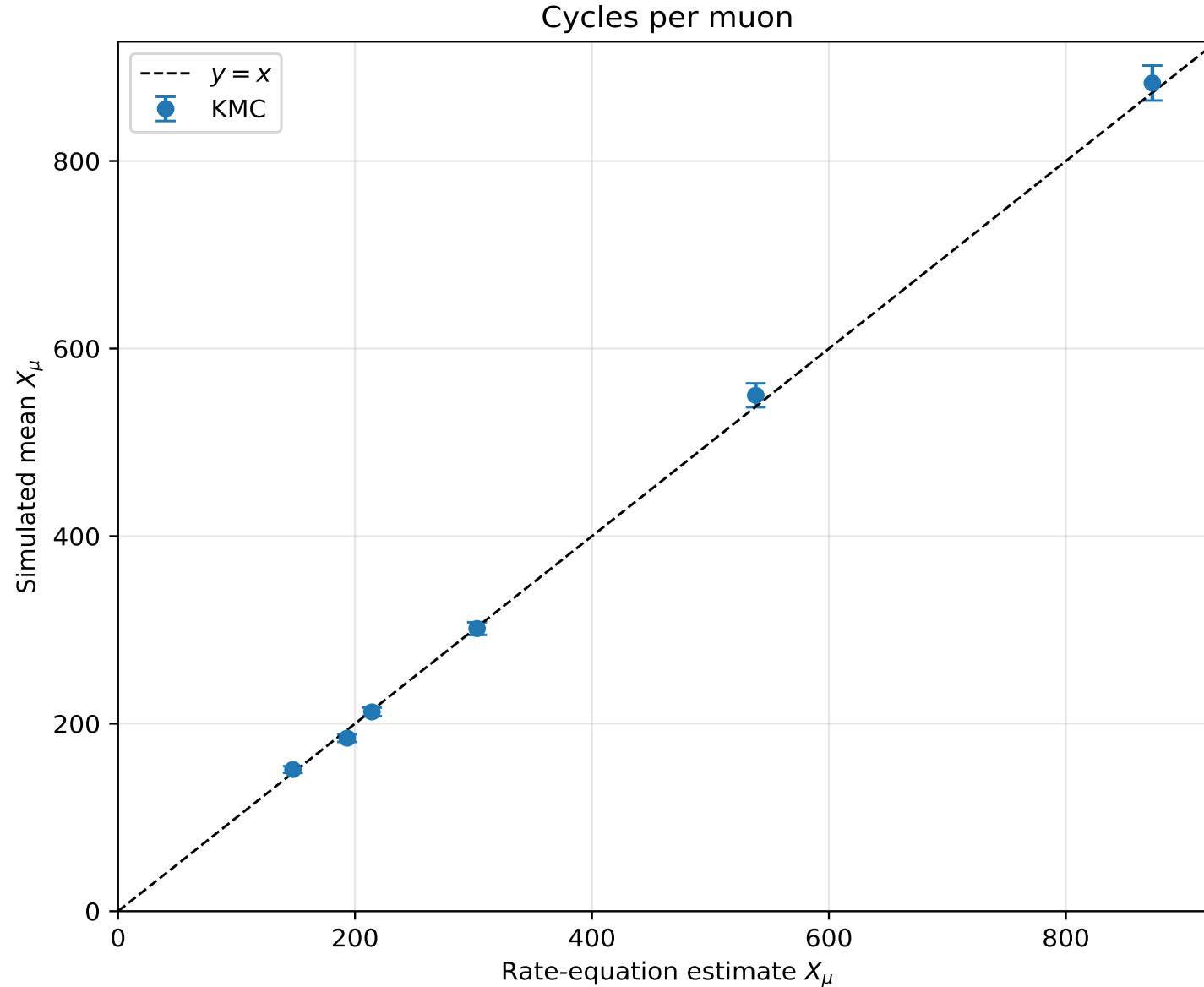
One muon cycles until decay or alpha sticking stops it.

$$X_{\mu} = \frac{1}{\omega_s + \lambda_{\mu}/\lambda_c}$$

Symbol	Meaning
X_{μ}	Mean cycles per muon
λ_c	Cycling rate
λ_{μ}	Muon decay rate
ω_s	Sticking probability

Our Baseline Estimate

KMC Matches The Reduced Model

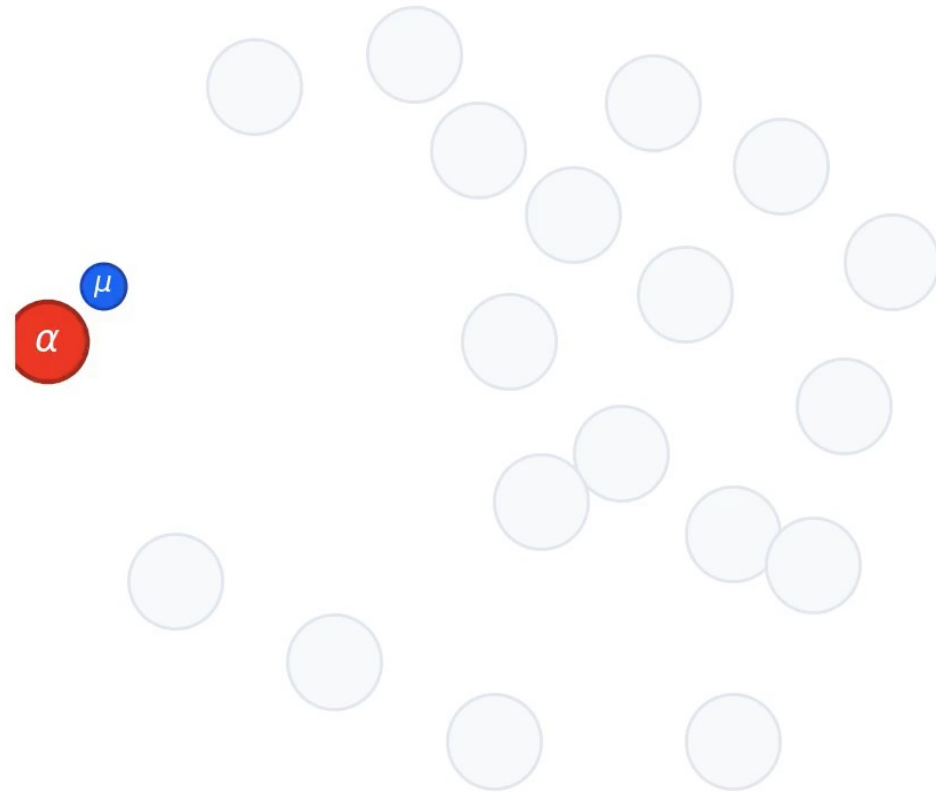


Reactivation Lowers Effective Sticking

- **Initial sticking:**
 μ captured by α
- **Reactivation:**
 μ can be stripped off again
- So only part of sticking is permanent

Initial Sticking ($\omega_s \approx 0.8\%$)

Muon stuck to the α after fusion



Effective Sticking Is A Shortcut

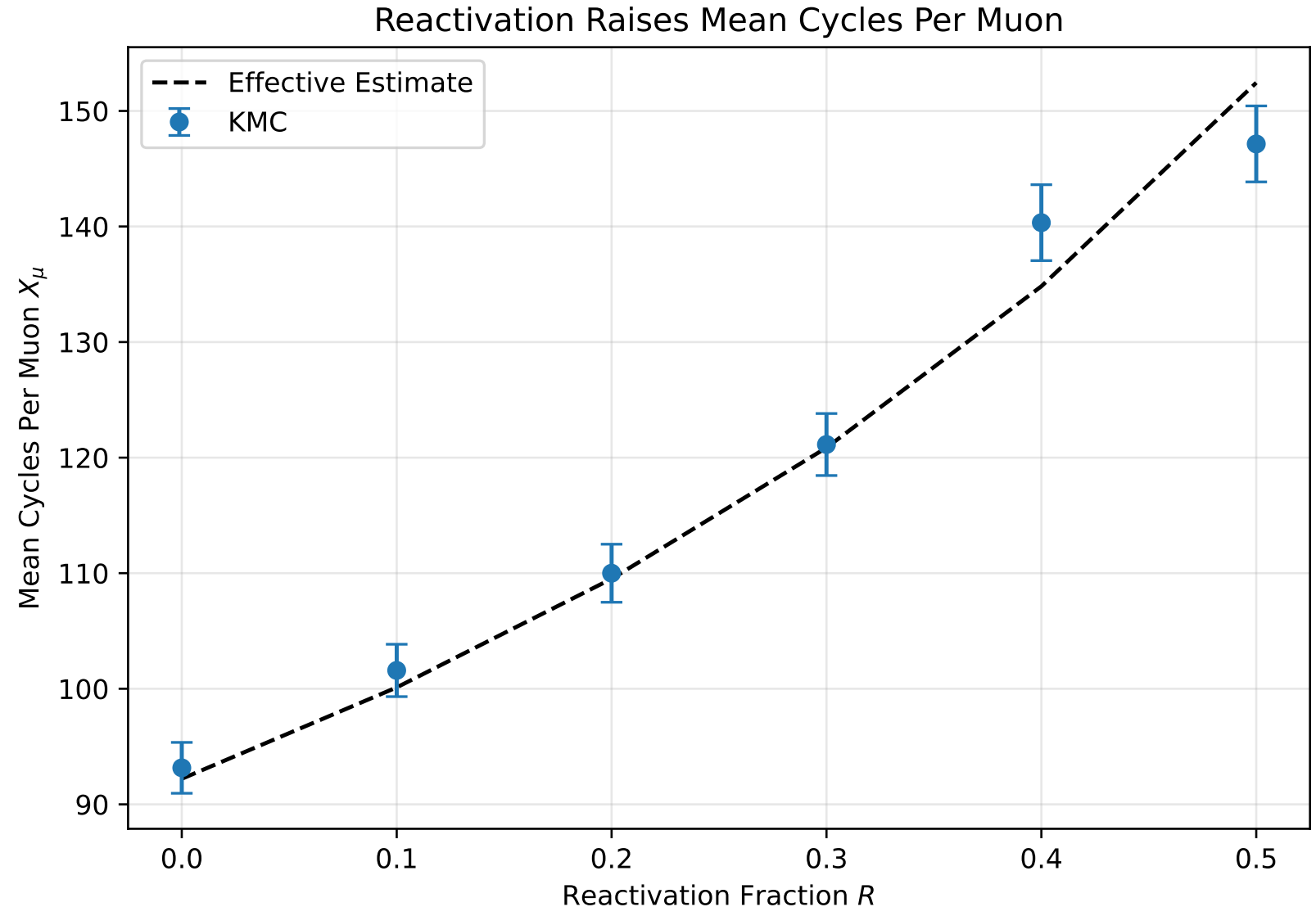
- Initial sticking happens after fusion
- R is the chance the muon returns
- Effective sticking is what remains permanent

$$\omega_S^{\text{eff}} = \omega_S^0 (1 - R)$$

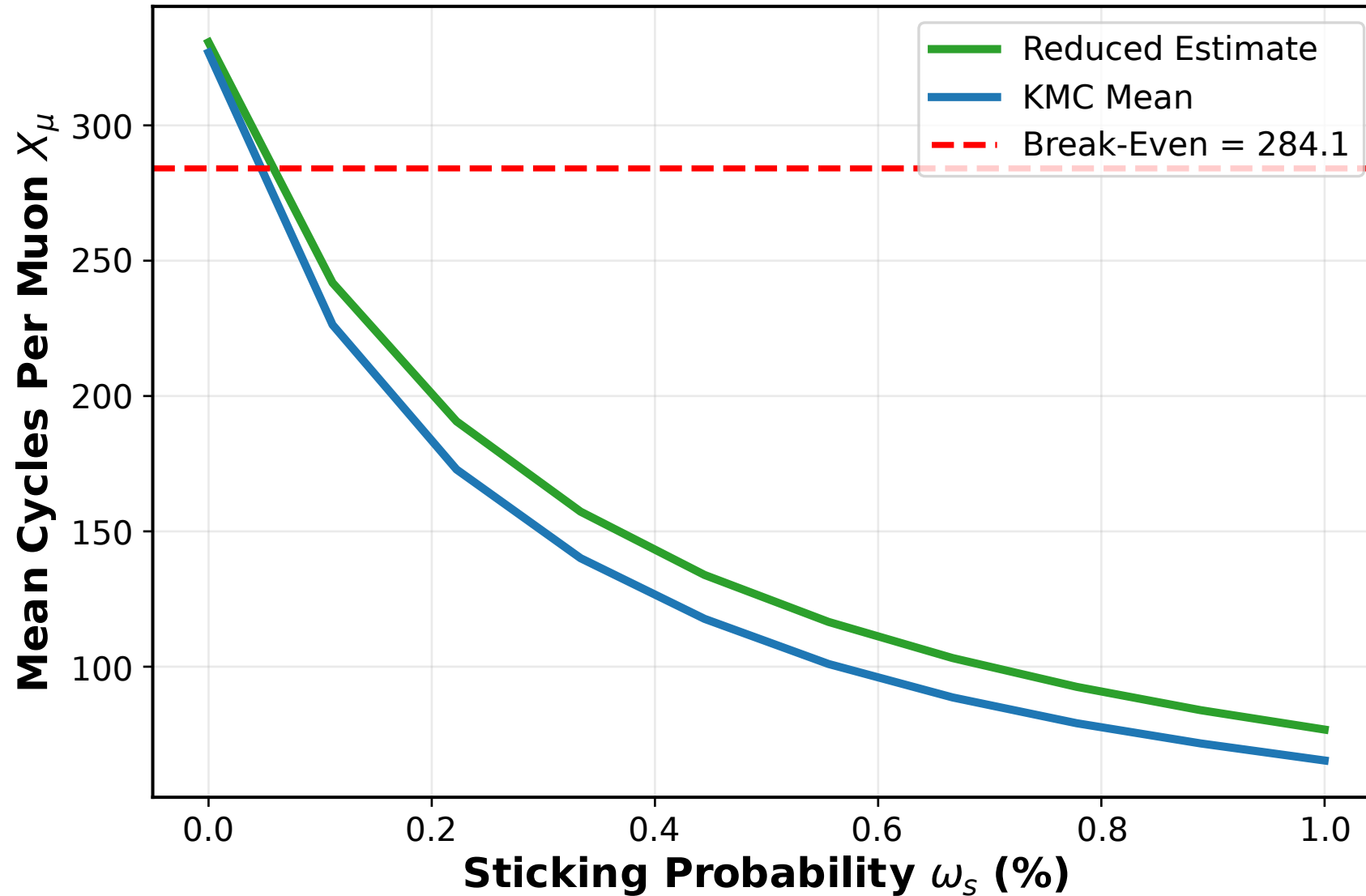
Symbol	Meaning
ω_S^{eff}	Effective Sticking Probability
ω_S^0	Initial Sticking Probability
R	Reactivation Coefficient

Reactivation Raises Mean Cycles Per Muon

- KMC With Explicit Recovery
- Dashed: Effective Sticking Estimate
- Recovery helps



Sticking Controls The Yield





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Fusions per muon

When do we break even?

How many fusions per muon are needed? (X_μ)

Energy per fusion

Energy to get muon

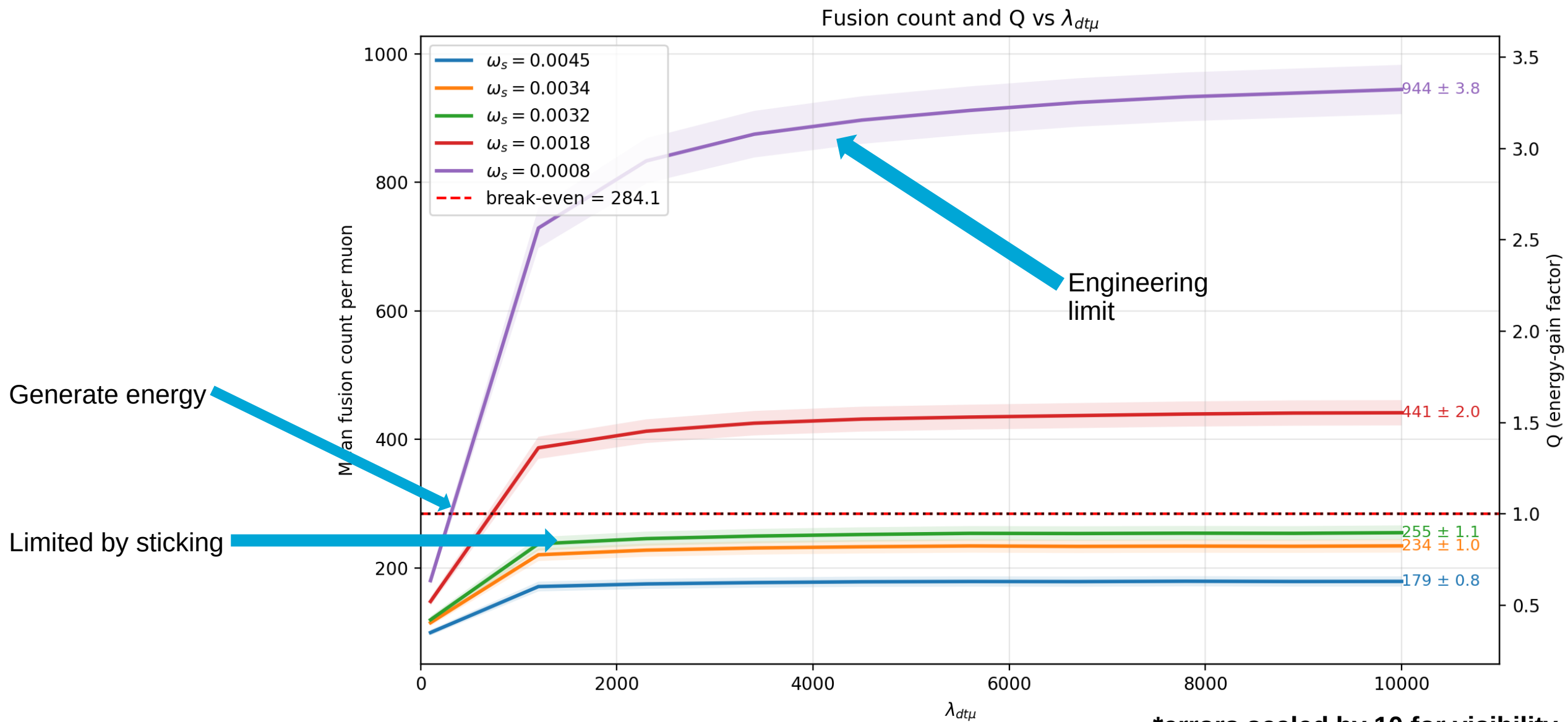
$$17.6MeV \times X_\mu - 5000MeV > 0 \Rightarrow X_\mu > 284.1$$

Fusion counts per muon

Energy-gain
factor

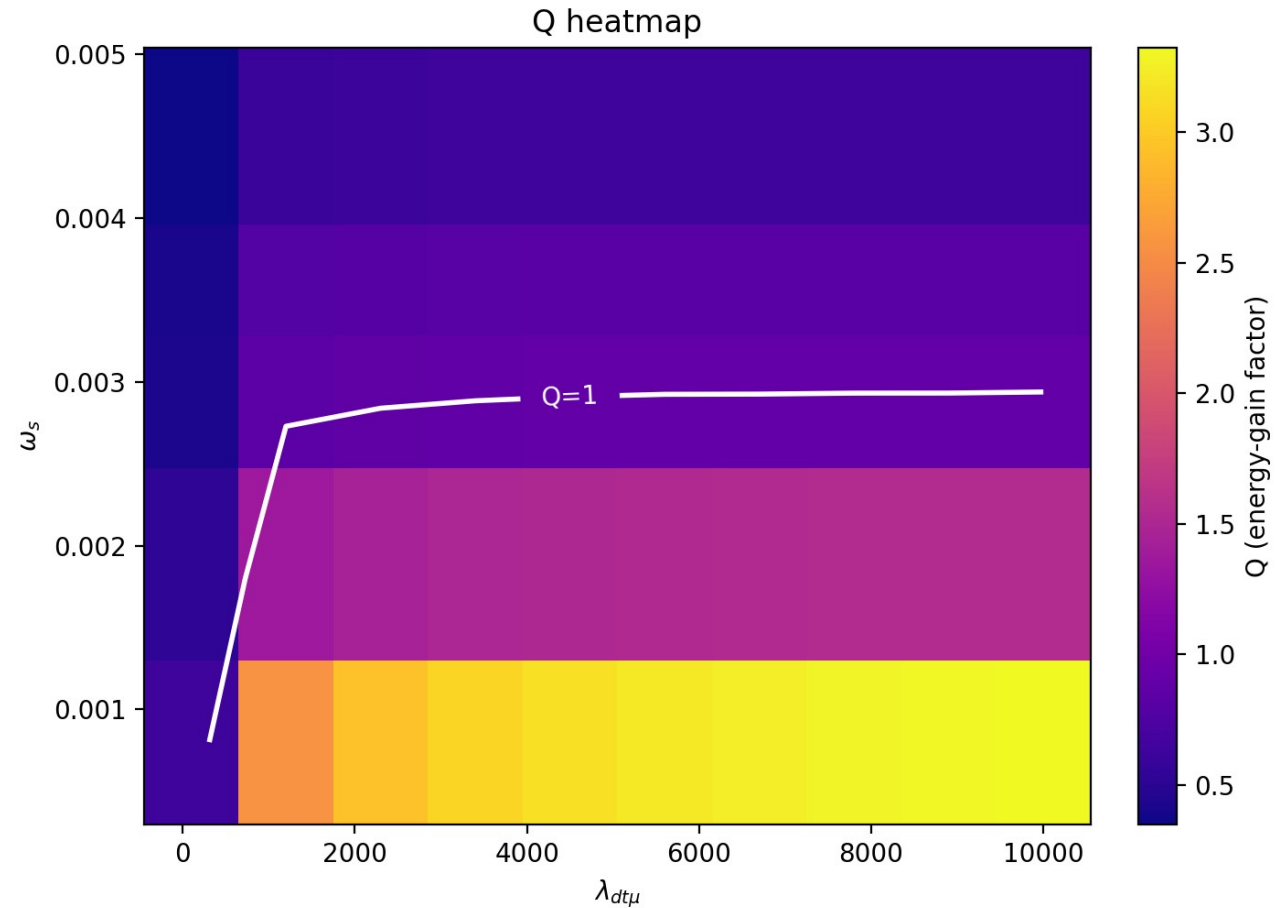
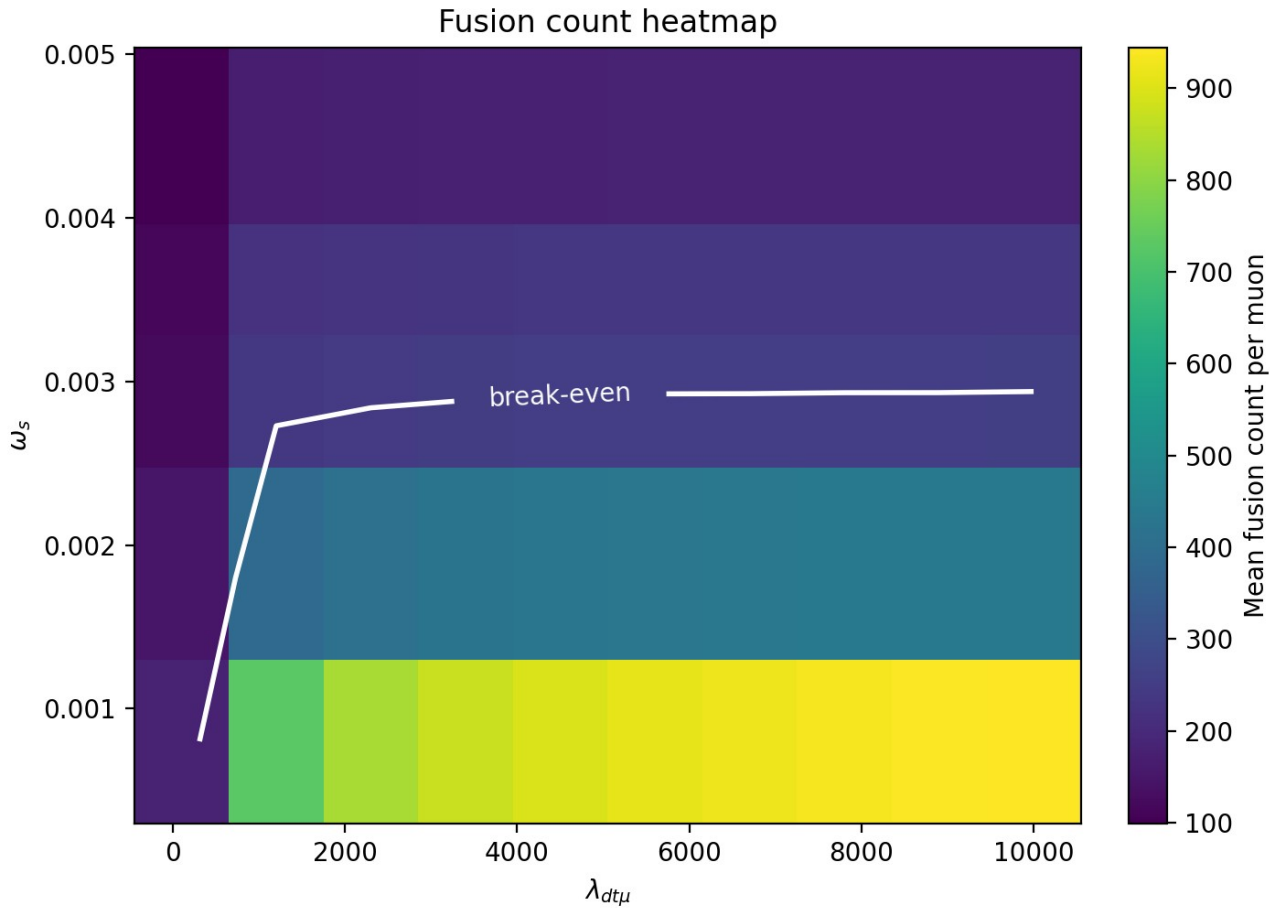
$$Q = \frac{17.6MeV \times X_\mu}{5000MeV} > 1$$

Molecular formation rate



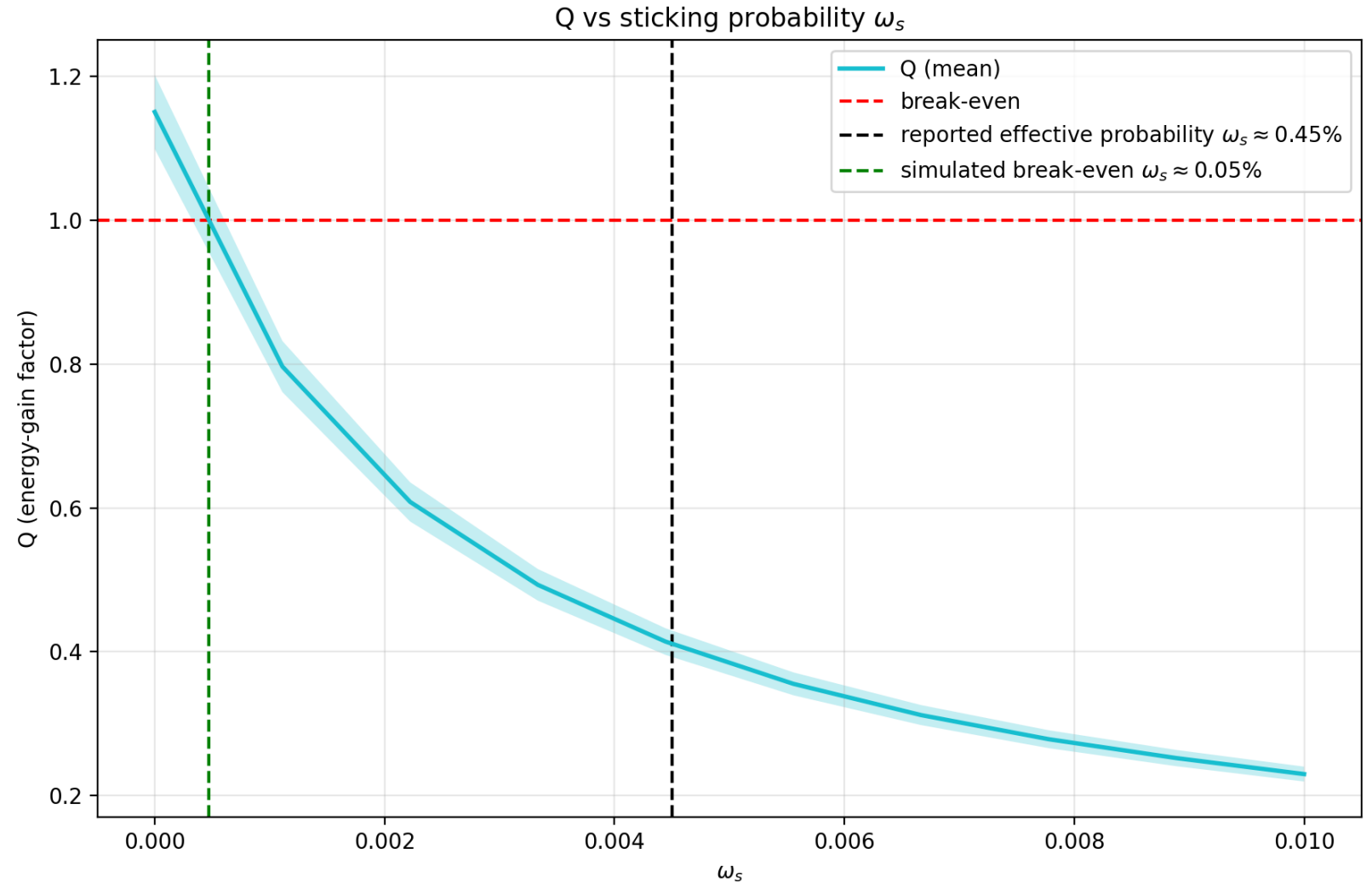
*errors scaled by 10 for visibility

Muonic Molecular Formation Rate



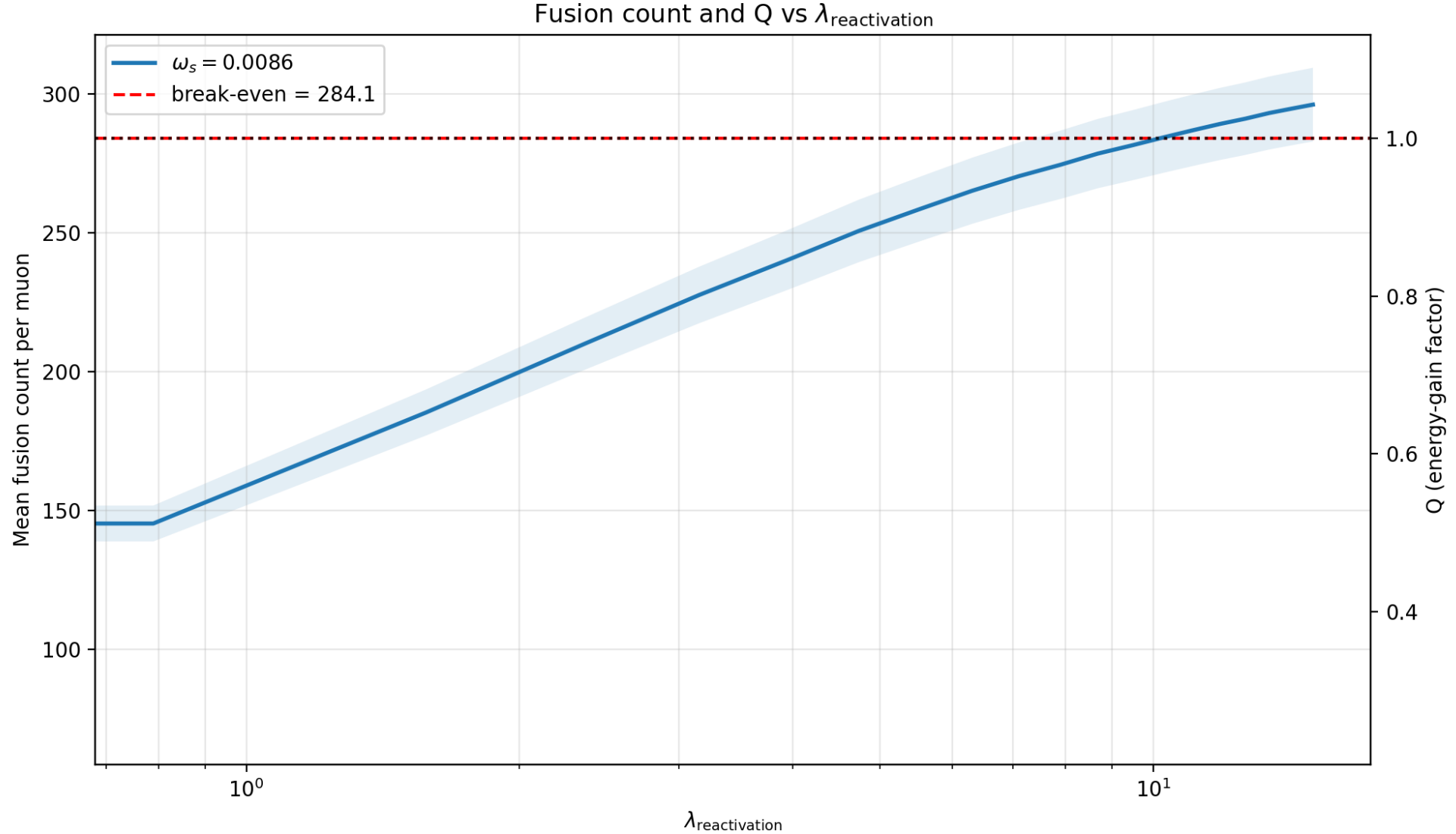
Sticking probability – the main bottleneck?

- Decreasing this alone is enough
- Realistic values are still too high



*errors scaled by 10 for visibility

Reactivation for realistic sticking

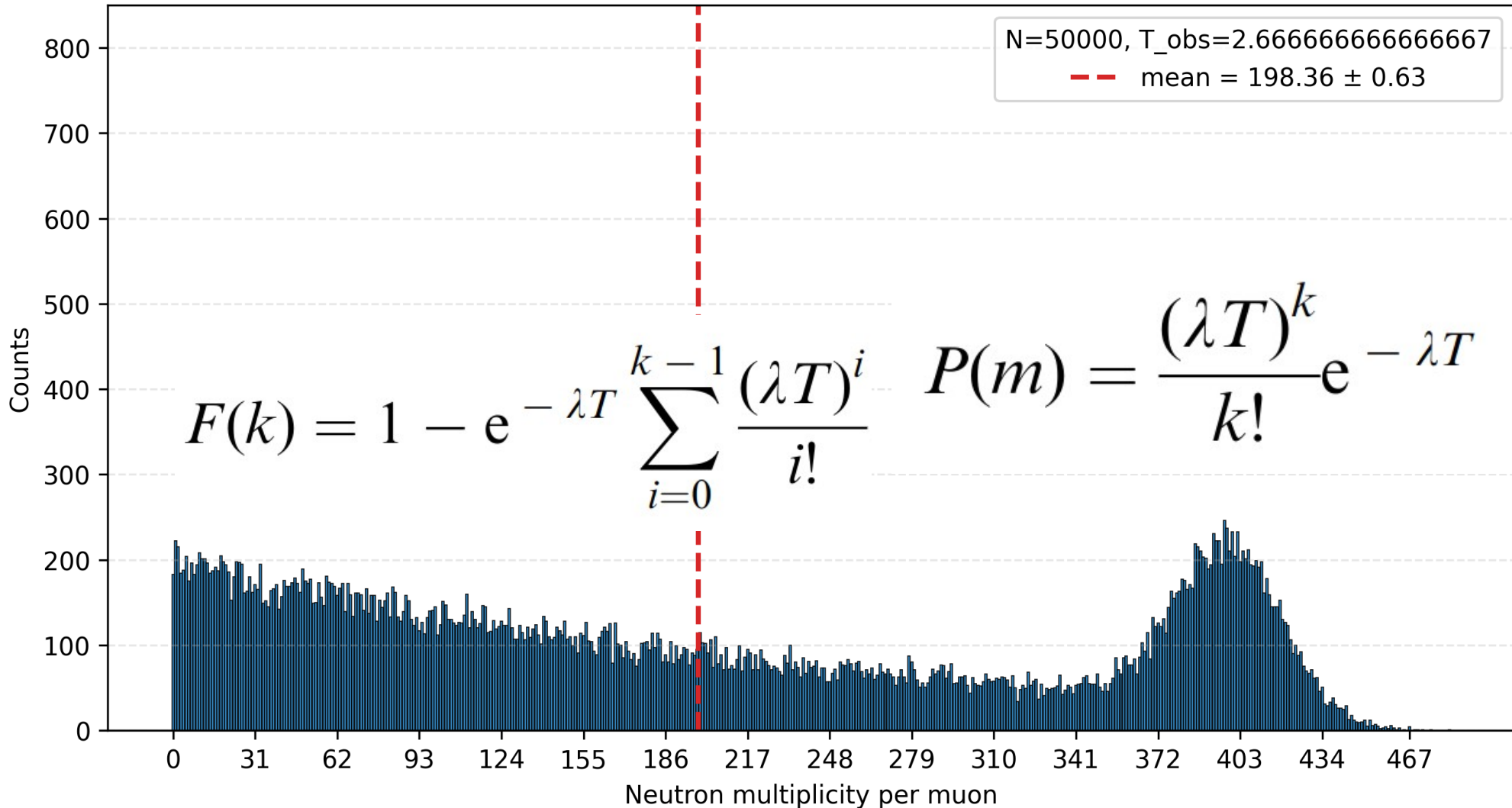


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Neutron multiplicity distribution

Muon-catalyzed fusion KMC: perfect-detection multiplicity counts

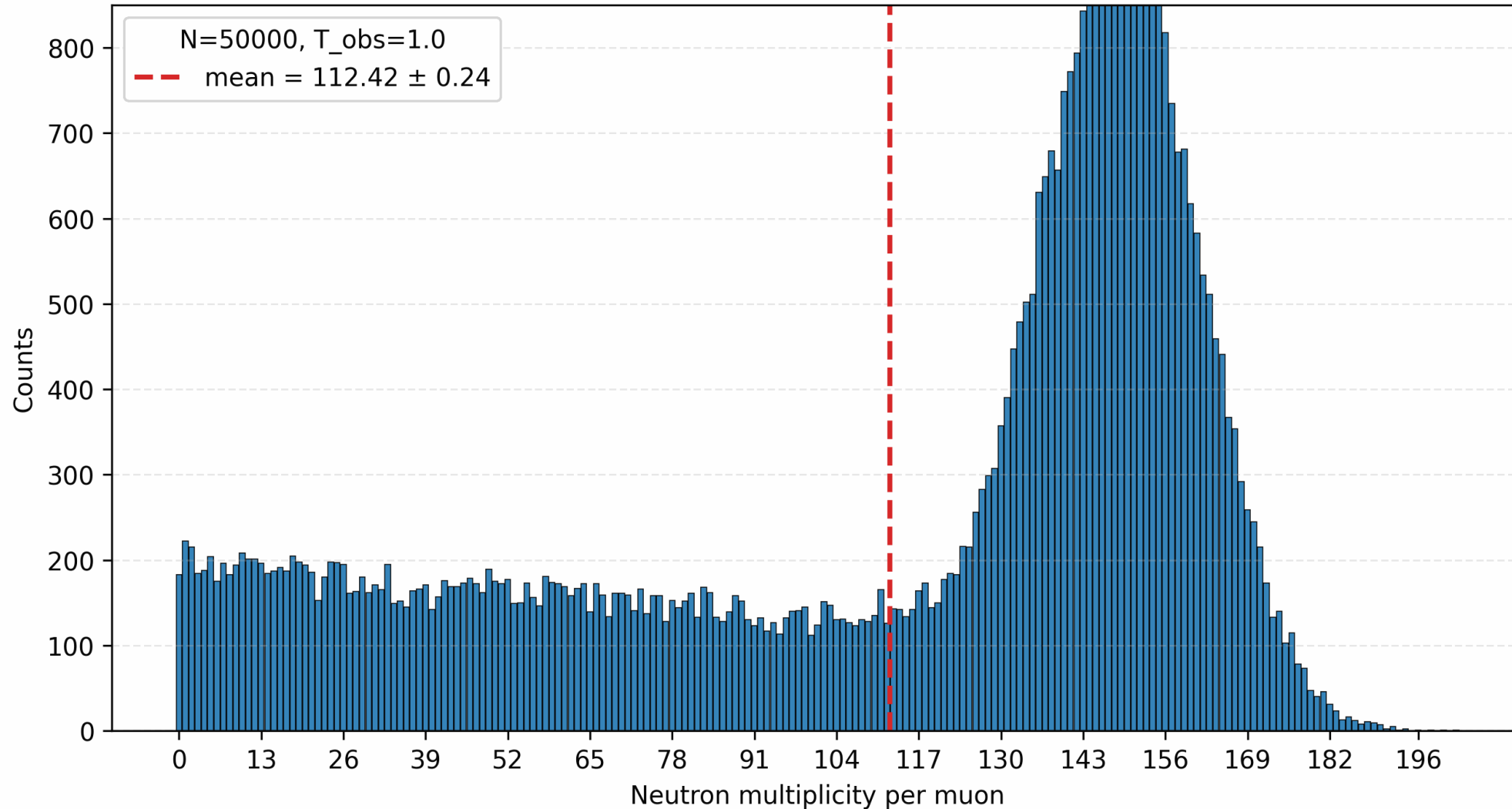
Neutron multiplicity count distribution



Neutron multiplicity distribution

Muon-catalyzed fusion KMC: perfect-detection multiplicity counts

Neutron multiplicity count distribution



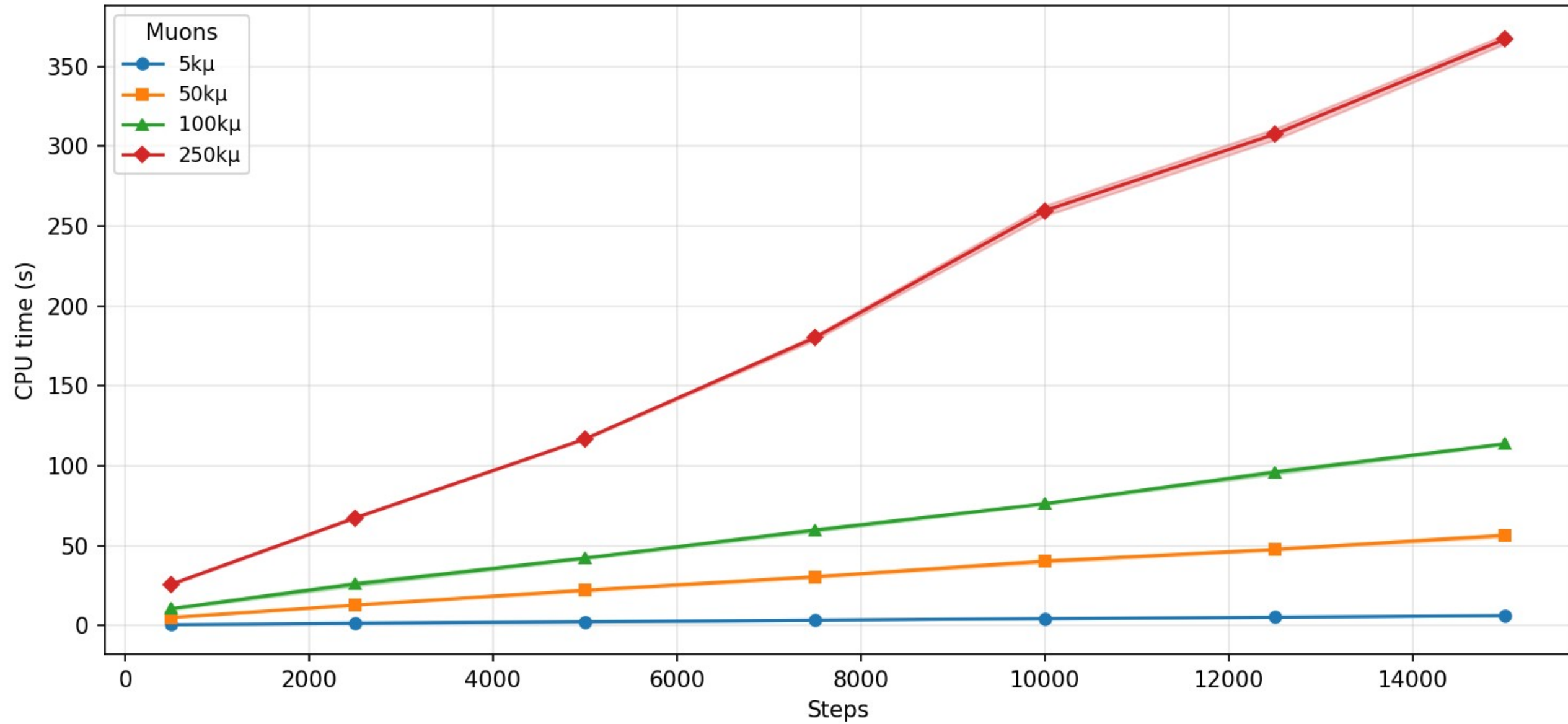
$\rightarrow X_\mu$



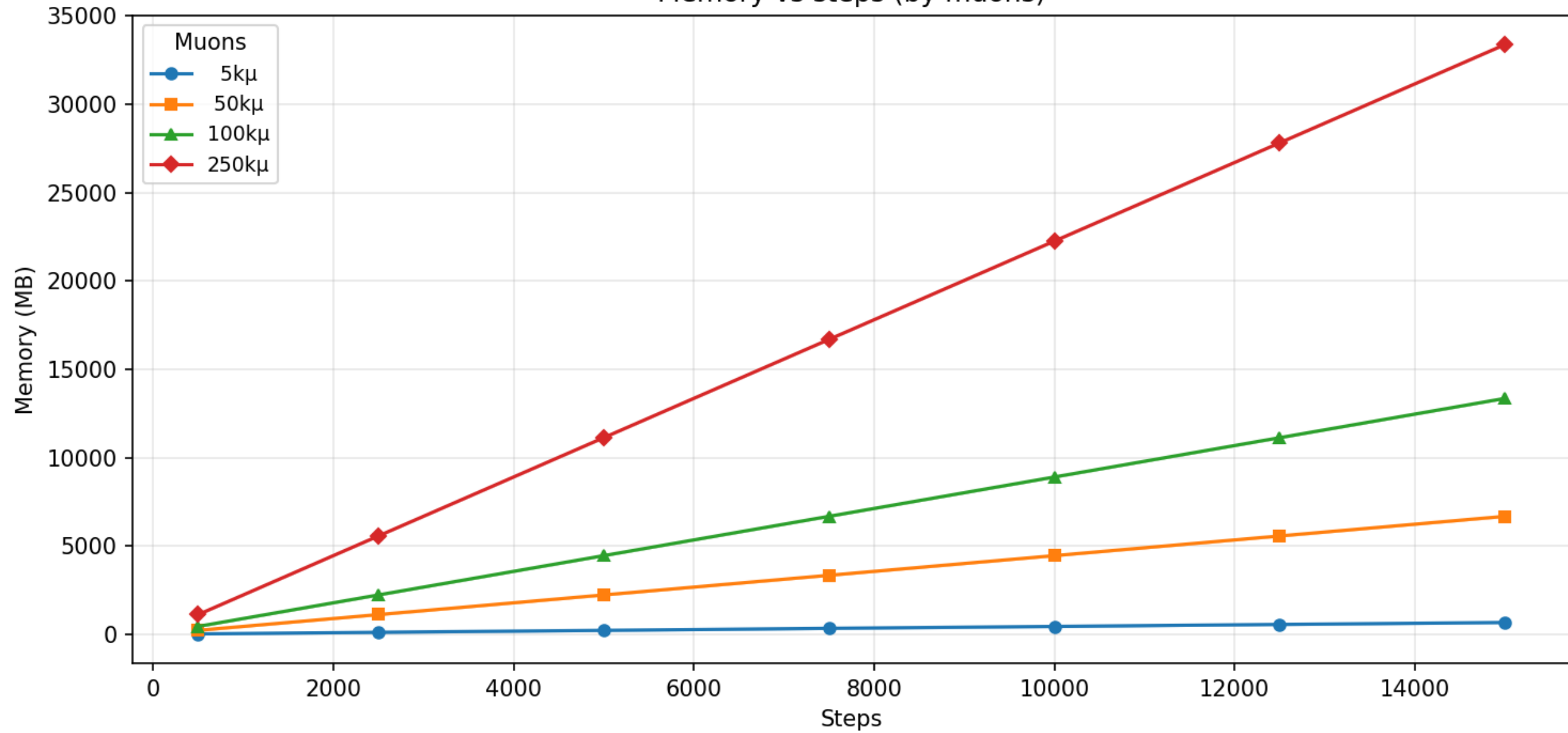
Performance

- Tested for Common Use Case
- Lenovo Legion Pro 5
- **CPU Heavy Implementation**
 - AMD Ryzen 9 7945HX
 - Clock: 3201 Mhz
 - 8 Cores
 - **32 GB RAM**
- Stress test on Idle System

CPU time vs steps (by muons)



Memory vs steps (by muons)



Conclusion

- KMC reproduces the reduced rate model
- Sticking is the main yield bottleneck.
- Molecule formation matters, but only after sticking is low enough
- Reactivation helps, but does not remove the break-even problem
- Break-even needs several bottlenecks improved together
- More complex physical modelling is most effective at the bottlenecks

Questions?